

TABLE I
VARIOUS TECHNOLOGIES
USED IN THE STRUCTURE
OF A SOLAR CELL

Technology	Efficiency at laboratory (per cent)
Organic cells	11.50
Dye-sensitised	11.90
Perovskite cells	21.00
Multi-crystal silicon	21.30
CdTe	21.50
CIGS	22.30
Single-crystal silicon	25.00
Ga As	27.50
Multi-junction (non concentrator)	38.80
Multi-junction (concentrator)	46.00

(Courtesy: www.synergyfiles.com)

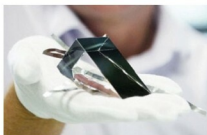


Fig. 5: Four-junction mini module embedded in a prism

tion at its most efficient wavelength, while the unused part of the light passes through to the next layer, and so on. Apart from the prism structure, logic remains the same as of a multi-junction solar cell.

Efficiency of a solar cell

Efficiency of a solar cell can be represented in many ways, namely, thermodynamic, charge carrier separation, quantum and conductance. Several factors and parameters are difficult to measure directly and, hence, other parameters are measured to calculate quantum efficiency.

Quantum efficiency can be measured in terms of internal quantum efficiency (IQE) and external quantum efficiency (EQE). IQE is the ratio of the number of charge carriers

TABLE II
MANUFACTURERS, MODELS, TYPE AND EFFICIENCY
OF SOLAR CELLS

Manufacturers	Model	Type	Module efficiency (per cent)
Spectrolab	XTJ Prime	Next triple junction	30.7
	XTJ	Next triple junction	29.5
	UTJ	Ultra-triple junction	28.3
SunPower	245-235	Monocrystalline	20.1
SunEdison	Silvantis R-Series	Monocrystalline	17.7
Yingli Solar	YL225C-24b	Monocrystalline	17.1
REC	Twinpeak Series	Monocrystalline	16.7
Ja Solar	JAM6 48-220/SI	Monocrystalline	16.6
Suntech	STP265/WEM	Polycrystalline	16.3
Trina	PA05	Polycrystalline	16.2
Jinko Solar	JKM265P	Multicrystalline	16.2
Canadian Solar	CS6P	Polycrystalline	15.9
Yingli Solar	YL205P-23b	Multicrystalline	15.8
Solar World	SW 260	Monocrystalline	15.5
Sharp	NU-U23F1	Monocrystalline	14.4
Yingli Solar	YL 235 P-29b	Polycrystalline	14.4
Kyocera	KD210GH-2PU	Polycrystalline	14.1

(Courtesy: www.theecexperts.co.uk)

(free electrons and holes) collected by the solar cell to the number of photons of given energy that shine on the solar cell from outside and are absorbed by the cell.

EQE is the ratio of the number of charge carriers collected by the solar cell to the number of photons of given energy shining on the solar cell from outside (incident photons). IQE is always greater than EQE. Efficiency of a traditional solar cell can be in the range of 14 to 24 per cent.

A solar panel (that converts sunlight to electric energy) and a thermal panel (that converts sunlight to heat energy used to produce hot water) are completely different. The latter is designed with an efficiency of up to 90 per cent.

Efficiency of a solar cell also depends on parameters like area of the cell, intensity of sunlight, wavelength of the photons, ambient temperature and so on.

Typically, efficiency is represented as energy at output/area of cell/time; for instance, 4.5 kWh/m²/day.

This means that a solar panel of 1m² size generates 4.5kWh electric energy per day. It may be more or less depending on environmental conditions. Tables I and II give show the different technologies used to make solar cells, models, types, manufacturers and efficiencies.

Types of solar cell

Solar cells can be broadly classified as:

Monocrystalline. Made up of a single crystal of silicon

Polycrystalline. Made up of many small crystals of silicon

Amorphous or thin-film. Made up of non-crystalline structure of silicon; has four types, namely, CIGS, Cdte, amorphous si and polymer organic

Advantages of solar cells

Main advantages of using solar panels to generate electricity are:

- A solar plant is a renewable power source and is not harmful to the environment.
- It is easy to operate and cost-